

Gravity is not a Pairwise Local Classical Channel

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It is currently believed that we have no experimental evidence on gravity-inspired modifications to quantum mechanics, such as the Diosi-Penrose model. Furthermore, it is widely accepted that the most auspicious approach to verifying such models are quantum tests with large massive systems, realized with optomechanical or large-molecule interferometric setups. Here we show that single-atom interference experiments achieving large spatial superpositions rule out the gravitational decoherence model of Kafri, Taylor and Milburn. Experiments thus show that gravitational interactions cannot be described as pairwise local classical channels between massive particles. We discuss how the same experiments impose constraints on other related models.

Introduction There is overwhelming experimental evidence that properties of local physical systems are incompatible with a classical description [1–8]. Nevertheless the possibility that gravity remains classical at a fundamental level is considered viable or even necessary [9–17]. Credence to such a hypothesis is supported by the absence of direct observations of quantum gravitational phenomena [18], anticipation of pernicious tensions between the foundational principles of quantum theory and general relativity [15, 16, 19] (see e.g. [20] for opposing views), and lack of a complete framework for quantum gravity [21].

The hypothesis of fundamentally classical gravity leads to models incorporating mechanisms suppressing coherence of spatial superpositions of “sufficiently massive” objects (which could otherwise produce non-classical states of the gravitational field at macroscopic scales). Such models can also be seen as attempts to explain how classical behaviour, in particular definite measurement outcomes, arises from the laws of quantum mechanics: the models enforce classicality of the centre of mass (CM) of macroscopic objects, where classicality is understood as lack of coherent spatial superpositions and macroscopicity is related to the strength of the gravitational field produced by the system [12, 22–24].

Measurement-and-feedback models of gravity such as those of Kafri, Taylor and Milburn (KTM) [25–27] also aim at enforcing classicality of gravitational interactions, but from an information-theoretic perspective, where classicality is understood as inability of the resulting channel to increase entanglement. Such an approach also predicts decoherence of the CM of gravitationally interacting systems in the position basis. At present, the KTM model is considered equivalent [14, 26, 28] to the Diosi-Penrose (DP) model [12, 13, 15, 24].

Here we show that the KTM model is incompatible with the results of atomic fountain interference experiments [29, 30]. We also clarify that it fundamentally differs from the DP

model: the quantities defining decoherence rates have different physical meaning and the models disagree about what states are affected and how strong is the decoherence.

Disproving these two types of models has also vastly disparate implications. Testing DP or Schrödinger-Newton theories [31, 32] would imply (in)validity of certain non-linear modifications to quantum mechanics whose scale is linked to the strength of gravity. Rebutting the KTM model, however, tells us something profound about the nature of the gravitational interaction itself: it cannot be modelled by local operations and classical communication (LOCC) (that is by an LOCC channel acting pairwise between massive particles).

Effective gravity from continuous measurements and feedback The KTM model of gravity as a classical channel is a particular case of pairwise continuous measurement with feedback [33, 34]: essentially it is a collisional decoherence model where a system interacts with a Markovian environment. Depending on the form and the strength of the interactions, and the state of the environment, collisional models can give rise to system dynamics ranging from exact unitarity under an effective potential to arbitrarily fast decoherence [35].

We now summarize the key aspects of the KTM model [26] (for more detail see appendix A). Consider a pair of particles that interact with a set of ancilla (environment). The interactions are local and comprise continuous measurement of each particle’s position with strength D and a feedback interaction with strength K . The measurement outcomes determine the magnitude of the subsequent feedback, which is a shift of the particles’ positions. The ancillae are assumed inaccessible; the scenario is thus equivalent to averaging over definite but unknown measurement outcomes and correspondingly applied feedback-shifts. It thus defines an LOCC channel [36] acting between the pair of masses. The resulting dynamics of the particles reads

$$\dot{\rho}_{12} = -\frac{i}{\hbar}[\hat{H}_0 + V_0, \rho_s] - \left(\frac{1}{4D} + \frac{K^2 D}{4\hbar^2} \right) \sum_{i=1}^2 [\hat{x}_i, [\hat{x}_i, \rho_s]], \quad (1)$$

where ρ_{12} is the state of both particles and $\hat{x}_1, \hat{x}_2$ is the displacement from the initial position of the respective particle. Apart from the non-unitary terms eq. (1) contains an effective

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tive unitary interaction $V_0 = K\hat{x}_1\hat{x}_2 +$ (local terms in $\hat{x}_1, \hat{x}_2$), which for $K := 2\frac{Gm_1m_2}{d^3}$ approximates the Newtonian potential between two masses m_1, m_2 at a distance $d + x_1 + x_2$: $V_0 \approx -G\frac{m_1m_2}{|d+x_1+x_2|}$, up to second order in x_1, x_2 .

The non-unitary terms describe decoherence in the position basis. The magnitude of the off-diagonal elements x_1, x_2 for each particle decays at a rate $\Gamma_{\text{KTM}} = \left(\frac{1}{4D} + \frac{K^2D}{4\hbar^2}\right)\Delta x^2$, where $\Delta x = |x_1 - x_2|$ is the “superposition size”. Importantly, Γ_{KTM} has a non-vanishing lower bound $\propto \frac{K}{2\hbar}$. The KTM model is thus essentially parameter-free. The resulting decoherence is fully characterised by the gradient of the force induced between the particles by the feedback – here fixed to reproduce Newton’s force between two point masses – and by the superposition size:

$$\Gamma_{\text{KTM}}^{\text{min}} = \frac{K}{2\hbar}\Delta x^2. \quad (2)$$

The effective interactions are necessarily accompanied by decoherence of at least the same strength, since LOCC channels are entanglement non-increasing [36] (see also ref. [35] for an explicit discussion).

Composite systems Here we summarize the main results for extending the KTM model to macroscopic systems (see appendix B). Let s_1, s_2 be rigid bodies with total masses M_1, M_2 , comprising N_1, N_2 elementary constituents, respectively, with masses $m_i, i = 1, \dots, N_1 + N_2$. We take the minimum (as in eq. (2)) of the decoherence rates for each pair whose evolution is described by (1). Moreover, we consider s_1 to be a test mass, in superposition of locations at a distance Δx while s_2 describes all the remaining matter (earth, ~ 500 kg of tungsten, etc) and is thus considered well localised. The resulting dynamics of the CM of s_1 reads

$$\dot{\rho}_{s_1} = -i[\hat{H}_0 + V, \rho_{s_1}] - \mathcal{D}_{\text{min}}[\hat{r}_1, [\hat{r}_1, \rho_{s_1 s_2}]], \quad (3)$$

where $V \approx -G\frac{M_1M_2}{|r_1-r_2|}$, with r_k the CM position of s_k and

$$\mathcal{D}_{\text{min}} := \frac{1}{2\hbar} \left(\sum_{i \neq j \in s_1} |K_{ij}| + \sum_{i \in s_1} \sum_{j \in s_2} |K_{ij}| \right), \quad (4)$$

where K_{ij} is the Newtonian force gradient between masses m_i, m_j in three dimensions. The corresponding minimal decoherence rate reads

$$\tilde{\Gamma}_{\text{KTM}}^{\text{min}} = \mathcal{D}_{\text{min}}\Delta x^2. \quad (5)$$

The sum of the unitary contributions approximates Newtonian interactions between all constituents – which is simply the gravitational potential energy between two point masses M_1, M_2 at a distance $|r_1 - r_2|$. However, the decoherence rate (4) in general differs from that for two point particles: first, it contains terms connecting constituents of s_1 (first sum) and s_1 with s_2 (second sum) and second, the rate (5) might be smaller than the rate given by the original KTM model (2) applied to the CMs of s_1, s_2 – for example when s_2 is a spherical shell of

matter with test particle s_1 at the centre¹. It can be shown (appendix B) that for an elementary test mass m near a surface of a homogeneous ball of mass M and radius R , the KTM decoherence rate is at least as large as

$$\tilde{\Gamma}_{\text{KTM}}^C = C \frac{GMm}{\hbar R^3} \Delta x^2, \quad (6)$$

with $C = 0.47$ for this particular geometry. For $C = 1$ the above reduces to the original KTM model applied to the CMs of the two systems.

Note that one cannot in general approximate the mass distributions to be continuous in this model. Contributions from the body’s own constituents diverge, and an explicit definition of the *fundamental* constituents is required. We propose that these should be the smallest constituents between which the binding energy contribution to the total mass can be neglected (since the model assumes that the total mass of the system is the sum of the masses of its constituents). We will consider atoms as such fundamental constituents when discussing macroscopic objects.

Comparison to Diosi-Penrose In the DP model the decoherence rate is given by a *self-interaction* between superposed amplitudes X, Y of a single system $\Gamma_{\text{DP}} = \frac{G}{2\hbar}[U(XX) + U(YY) - 2U(XY)]$, where $U(XY) = -G \int d^3r \int d^3r' \frac{f_X(r)f_Y(r')}{|r-r'|}$ is the self-interaction between mass-distributions f_X, f_Y associated with the superposed configurations X, Y of a *single system* [13]. In contrast, in the KTM model decoherence is given by the (measurement-induced) gravitational interaction between *different systems*. While both models predict decoherence in the position basis, and whose magnitude is related to gravity, they fundamentally differ both conceptually, Table I, and quantitatively, Table II as follows:

- 1) For a single point particle Γ_{DP} diverges and requires a cut-off δ in the *coherent spread* of the particle’s wave-function [13], while the KTM model is well-defined for point particles.
- 2) A single elementary particle in an otherwise empty universe decoheres in the DP model (arbitrarily fast, for arbitrary small δ), whereas it does not in the KTM model: If all masses are arbitrarily far from the test particle $\tilde{\Gamma}_{\text{KTM}}^{\text{min}} \rightarrow 0$. This is conceptually very desirable feature, since the location of a single particle in an empty universe has no physical meaning. (It is not possible to distinguish any such “two locations”, therefore, arguing that the particle is – or is not – in a superposition of these “two locations” also has no physical meaning.)
- 3) For a particle of mass m in a superposition Δx , at a distance d from a point mass M the KTM decoherence rate depends both on M and d . For fixed M : $0 < \Gamma_{\text{KTM}}^{\text{min}} < \frac{mc^2}{\hbar} \left(\frac{\Delta x}{R_s}\right)^2$ where the lower bound is for $d \rightarrow \infty$ and the upper is for

¹ Let s_2 to be a spherical shell of radius r made of N particles each of mass m , and s_1 – an elementary particle inside the shell. Sum over the constituents of the shell, eq. (4), yields a finite decoherence rate, which can be made arbitrarily small by increasing $r \rightarrow \infty$. However, decoherence predicted by the model applied directly to the CM of s_2 , eq. (2), is arbitrarily large for s_1 arbitrarily close to the shell’s centre, and is independent of r .

TABLE I. General form for spherical mass distributions of the decoherence rates Γ_{DP} (Diosi-Penrose) and $\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$ (Kafri-Taylor-Milburn) (5) [13, 37]. δ denotes coherent cut-off of DP model, and Δx is the superposition size. Case (iii) considers decoherence of the CM of a body comprising N_1 constituents of mass m in the presence of another body comprising N_2 masses m . Whereas $\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$ depends on the gravitational force gradients between *different* particles, Γ_{DP} depends on the *self-interaction* between superposed amplitudes of the same particle.

Scenario:	Γ_{DP}	$\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$
(i) single particle mass m	$\frac{Gm^2\Delta x^2}{2\delta^3\hbar}$ for $\delta \gg \Delta x$ $\frac{2Gm^2}{\delta\hbar} \left(\frac{6}{5} - \frac{\delta}{\Delta x}\right)$ for $\delta \ll \Delta x$	0
(ii) two particles masses m, M ; distance d	same as (i)	$\frac{GmM\Delta x^2}{d^3\hbar} = \begin{cases} 0, & d \rightarrow \infty \\ \frac{mc^2}{\hbar} \left(\frac{\Delta x}{R_S}\right)^2, & d \rightarrow R_S := \frac{2GM}{c^2} \end{cases}$
(iii) two composite bodies; masses N_1m, N_2m ; dist. d_{ij}	$N_1 \frac{Gm^2\Delta x^2}{2\delta^3\hbar}$; $\delta \gg \Delta x$ $N_1 \frac{2Gm^2}{\delta\hbar} \left(\frac{6}{5} - \frac{\delta}{\Delta x}\right)$ for $\delta \ll \Delta x$	$\frac{Gm^2\Delta x^2}{\hbar} \left(\sum_{i \neq j=1}^{N_1} K_{ij} + \sum_{i=1}^{N_1} \sum_{j=N_1+1}^{N_1+N_2} K_{ij} \right)$

TABLE II. Comparison of decoherence times $1/\Gamma_{\text{DP}}$ and $1/\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$ predicted by Diosi-Penrose and Kafri-Taylor-Milburn models for matter-wave interference experiments [29, 38–40]. For all tests $1/\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$ contains the contribution from earth $M_{\oplus} \sim 6 \times 10^{24}$ kg, $R_{\oplus} \sim 6 \times 10^3$ km, and for experiment [38] additionally from 24 bars of tungsten^a each with mass ~ 21.5 kg, and 6 of each at approximate distances of 107.6 mm, 177.6 mm, 279.5 mm and 313.1 mm from the atoms (C configuration). For Γ_{DP} we took $\delta = 10^{-15}$. The column labels denote: m – the mass of the interfering particles; M – “source” mass/masses; d – distance between the source and the test mass (both relevant only for Γ_{KTM}); Δx – the superposition size. For simplicity we treat all test masses as single particles. The KTM decoherence time $\sim 10^{-3}$ s (first row) is much shorter than 2 s timescale of the experiment [29], which we analyse in detail further in this work.

Experiment	m [Kg]	M [Kg]	d [m]	Δx [m]	$1/\Gamma_{\text{DP}}$ [s]	$1/\tilde{\Gamma}_{\text{KTM}}^{\text{min}}$ [s]
10 m atomic fountain with ^{87}Rb [29]	1.4×10^{-25}	$M_{\oplus}$	$R_{\oplus}$	0.54	3×10^{10}	2×10^{-3}
two atomic fountains with ^{87}Rb [38] (operating as gravity-gradiometer)	1.4×10^{-25}	$M_{\oplus}$ 4×129	$R_{\oplus}$ 0.11, 0.18, 0.28, 0.31	1.86×10^{-3}	3×10^{10}	2×10^1
large-molecule interferometry [39]	1.6×10^{-23}	$M_{\oplus}$	$R_{\oplus}$	2.7×10^{-7}	3×10^6	6×10^7
PcH ₂ diffraction on alga skeleton [40]	8.2×10^{-25}	$M_{\oplus}$	$R_{\oplus}$	2×10^{-7}	1×10^9	2×10^9

^a Tungsten bars were used to induce controllable gravity gradient between two atom interferometers operating as gravity gradiometer.

$d = R_S = 2GM/c^2$ (the Schwarzschild radius of M). In contrast, Γ_{DP} is entirely independent of the particle’s gravitational environment.

4) The KTM model requires a cut-off in the minimum distance between elementary masses (and not their coherent spread, as in DP). Moreover, the KTM “self interaction” terms – first sum in eq. (4) – are purely classical. They connect *different constituents* of a composite system – not different points of a single system wave-function.

5) Finally, the KTM model predicts vanishing decoherence when all force gradients K_{ij} are negligible. Importantly, the linear terms in particles’ positions can remain finite and the *sum of the homogeneous potential contributions* can be induced without decoherence. The KTM model is thus compatible (albeit to such limited extent) with the equivalence principle – since it does not predict any decoherence for an accelerating particle.

KTM vs Atomic Fountains Below we test the KTM model against experiments using atom interferometers with large

momentum transfer (LMT) [29, 30]. We treat the interfering atom as a test mass s_1 and earth as the massive ball s_2 in (6).

In LMT interferometers a sequence of $N \frac{\pi}{2}$ -pulses from a laser implements a beam splitter, preparing the atoms in a superposition of wave packets with momentum difference $2N\hbar k$ in the vertical direction, where k is the laser wave-number. For time T the wave packets spatially separate; then a sequence of π -pulses exchanges their momenta and at time $2T$ the wave packets interfere at a final beam splitter ($N \frac{\pi}{2}$ -pulses). For an atom of mass m the vertical separation between the wave packets is $\Delta x(t) = 2N\hbar kt/m$ for $t < T$ and then symmetrically decreases until $t = 2T$. Integrating $\tilde{\Gamma}_{\text{KTM}}^{\text{C}}$ along the paths separated by $\Delta x(t)$, we obtain maximum visibility predicted by the KTM theory

$$V_{\text{KTM}}^{\text{max}} = e^{-\frac{2}{3}C \frac{G\hbar M_{\oplus}}{mR_{\oplus}^3} (2Nk)^2 T^3}, \quad (7)$$

We compare in Fig 1 this prediction against measured visibilities [29] and [30]. The relevant parameters are $M_{\oplus} = 6 \cdot 10^{24}$

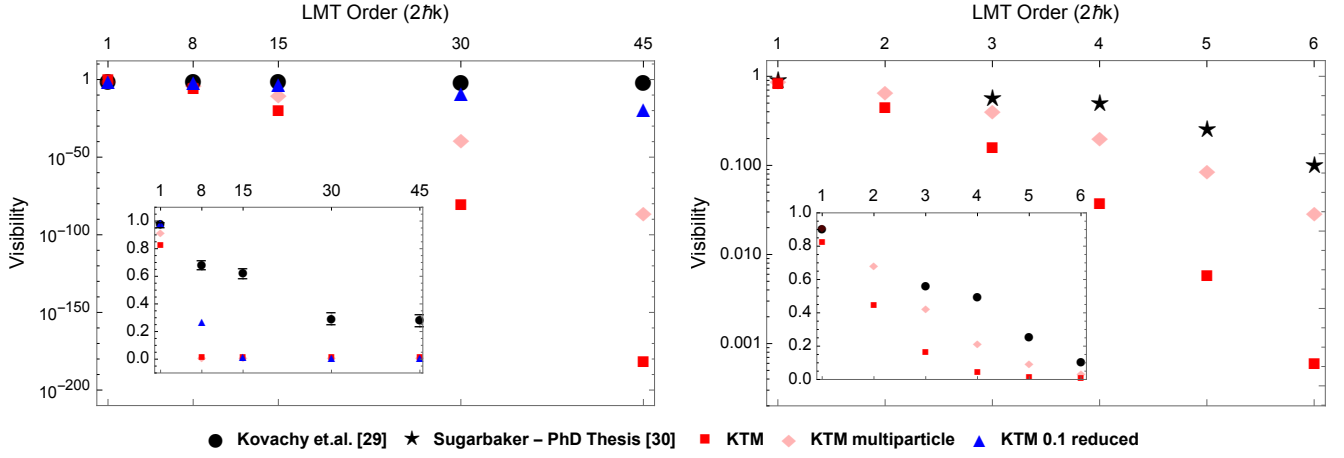


FIG. 1. Comparison of the visibility predicted by the KTM model eq. (7) with $C = 1$ (red squares), its multi particle correction with $C = 0.47$ (pink diamonds), and a reduced KTM correction with $C = 0.1$ (blue triangles) with the visibilities measured in two atom fountain experiments: [29] (black dots) in the left figure and [30] (black stars) in the right figure, both in a logarithmic scale and as a function of the LMT order. The insets correspond to the same data but in a linear scale. In the case of [29], the errors reported are included. The local classical channel model, even with the KTM reduction, is incompatible with experimental data. In both experiments, large momentum transfer (LMT) of order $2N\hbar k$ for $N = \{1, 8, 15, 30, 45\}$ in [29] (left figure) and $N = \{1, 3, 4, 5, 6\}$ in [30] (right figure) was realized to implement an interferometer with ^{87}Rb atoms, resulting in superposition sizes of up to 8.2cm in [30] and up to 54cm in [29].

kg; $R_{\oplus} = 6 \cdot 10^3$ km, $\frac{\hbar k}{m} = 5.8 \frac{\text{mm}}{\text{s}}$, $m = 1.4 \cdot 10^{-25}$ kg (^{87}Rb); $T = 1.15$ s in ref. [30] and $T = 1.04$ s in ref. [29].

Both the original KTM model ($C = 1$) and the multi particle correction ($C = 0.47$) predict maximal visibilities that are below the measured ones by factors of $\gtrsim 1$ to $\sim 10^{180}$ (for large-area setups). Taking into account finite duration of light-atom interactions would introduce a correction to the path separation. In the insets of Fig. 1 we show that even if this resulted in a reduction of the decoherence rate to $C = 0.1$, the resulting visibilities would still be smaller than the measured ones by factors ranging from ~ 2.5 to $\sim 10^{18}$.

Discussion A dedicated test of the KTM model could be realised with atom-fountain gravity-gradiometer: two interferometers with vertical separation L and a large mass M (in the plane of, say, the lower interferometer), whose horizontal distance d to the atoms can be varied [38, 41, 42] (c.f. Table I). The KTM model predicts that a differential phase noise will be induced between the two interferometers as a function of d . With $M = 252$ kg and d varying between 0.25 and 0.5 m, an experiment with LMT order $10\hbar k$ and $T = 0.5$ would see the lower interferometer's contrast correspondingly varying between 0.5—0.65, while the upper – between 0.62—0.64. Setup of ref. [42] already used an 84 kg source mass – our proposed experiment requires using two additional such masses in this setup. Realisation of such an experiment would allow relaxing some of the assumptions, e.g. about the mass distribution of the earth.

It would be desirable to also conduct non-interferometric tests of the KTM model to ensure no systematic errors affect the conclusions. For an optomechanical experiment, in order to detect the KTM decoherence on top of the thermal noise, the mechanical frequency Ω , quality factor Q and the temperature T of the mechanical oscillator must satisfy $T\Omega/Q < G\hbar M_{\oplus}/2k_B R_{\oplus}^3 \sim 10^{-18}$ K/s, where k_B is the Boltzmann

constant; a state of the art setup [43] with $Q = 2 \times 10^7$, $\Omega/2\pi = 1$ Hz, $T = 4$ K yields $T\Omega/Q \sim 10^{-6}$ K/s. For the model to be discernible from the measurement noise, the measurement frequency ω must satisfy $\omega^2 < GM_{\oplus}/R_{\oplus}^3 \sim 10^{-6}$ Hz², whereas the value considered in [43] (at the standard quantum limit) gives $\omega^2 \sim 10^6$ Hz². Optomechanical sensitivities are thus not yet sufficient to test the KTM theory.

The model is also far from the sensitivities of torsion balance experiments. For setups with 1—10 kg masses the model yields $\Gamma_{\text{KTM}}^{\text{min}} \sim 10^{25}$ Hz, while experimental bounds, obtained from refs. [44, 45], is $\sim 10^{40}$ Hz (see appendix C).

Conclusions We have discussed a model of low-energy Newtonian gravity as local classical channels acting pairwise between elementary constituents of massive bodies. The resulting gravity is classical (it cannot create entanglement) at a fundamental level, as well as between CMs of composite systems. Crucially, it has been shown [25–27] that a channel giving rise to the KTM unitary term but with decoherence smaller than the KTM lower bound is already non-classical, with the capacity to create entanglement. This minimal required decoherence is, however, already stronger than decoherence observed in atom-fountain interferometers. The experiments thus strongly constrain local classical channel model of gravity and suggest that gravitational interactions in principle have the capacity to convey entanglement, at least between small masses.

Collisional models with the required capacity and scaling can be constructed e.g. by relaxing the assumption of local system-ancilla interactions [35, 46] or by minimizing instead the heating rate (equivalent to the amount of energy introduced to the systems because of the ancillae), which yields a decoherence rate smaller by a factor of $\sqrt{m/M}$. Empirically constraining these more general models remains an interesting subject for further investigation.

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Appendix A: Essentials of the KTM model

A collisional decoherence model with a time-scale τ describes evolution of the state as $\rho_s(t+\tau) = \text{Tr}_A\{\hat{U}(\tau)(\rho_s \otimes \rho_a)\hat{U}^\dagger(\tau)\}$, where ρ_s and ρ_a are the density matrices of the system and the environment (ancilla), respectively, and $\hat{U}(\tau)$ describes their joint unitary evolution; the trace is over the ancilla.

The KTM model as defined in ref. [26] considers a quasi one-dimensional case of two essentially point-like massive particles. At each step of the collisional dynamics the massive particles interact with ancillas a_1, a_2 via two interaction terms: the “measurement” interaction, described by $\hat{x}_1 \otimes \hat{p}_{a_1} + \hat{x}_2 \otimes \hat{p}_{a_2}$ and the “feedback” interaction $K\hat{x}_1 \otimes \hat{x}_{a_2} + K\hat{x}_2 \otimes \hat{x}_{a_1}$. Here $\hat{x}_i$, $i = 1, 2$ are the position operators of the i^{th} mass, and $\hat{x}_{a_j}, \hat{p}_{a_j}$, $j = 1, 2$ are position and momentum operators of the j^{th} ancilla. For Gaussian states of ancillas with width σ , and in the continuous-interaction limit ($\tau \rightarrow 0$) the following master equation results [33, 34, 47]:

$$\dot{\rho}_s = -\frac{i}{\hbar}[\hat{H}_0 + K\hat{x}_1\hat{x}_2, \rho_s] - \left(\frac{1}{4D} + \frac{K^2D}{4\hbar^2}\right) \sum_{i=1,2} [\hat{x}_i, [\hat{x}_i, \rho_s]],$$

where $D := \lim_{\tau \rightarrow 0, \sigma \rightarrow \infty} \tau\sigma$. For $K = 2\frac{Gm_1m_2}{d^3}$ and $\hat{H}_0$ describing two harmonic oscillators with masses m_1, m_2 , the unitary term $K\hat{x}_1\hat{x}_2$ induced by the interactions describes the normal mode splitting due to an effective Newtonian potential [26].

By adding to the feedback interaction local terms $-\frac{d^2K}{2}\left(\frac{1}{2} + \frac{\hat{x}_1+\hat{x}_2}{d} + \frac{\hat{x}_1^2+\hat{x}_2^2}{d^2}\right)$ acting trivially on the ancilla, the effective unitary term becomes

$$V_0 = -G\frac{m_1m_2}{d} \left(1 - \frac{x_1 + x_2}{d} + \left(\frac{x_1 + x_2}{d}\right)^2\right)$$

which approximates Newtonian gravitational potential $-G\frac{m_1m_2}{|X_1-X_2|}$ up to second order in x_1, x_2 ; $X_1 - X_2 \equiv d + x_1 + x_2$. This extension recovers the homogeneous part of the potential in addition to the force gradients and yields

$$\dot{\rho}_s = -\frac{i}{\hbar}[\hat{H}_0 + V_0, \rho_s] - \left(\frac{1}{4D} + \frac{K^2D}{4\hbar^2}\right) \sum_{i=1}^2 [\hat{x}_i, [\hat{x}_i, \rho_s]].$$

which is eq. (1).

Appendix B: II. KTM model for composite systems

Here we extend the KTM model to composite systems in three dimensions. We consider two systems s_1 and s_2 consisting of N_1 and N_2 elementary constituents (atoms), respectively, with masses m_i , $i = 1, \dots, N_1 + N_2$. The classical gravitational potential energy between any two constituents i, j reads

$$V_{ij} = \frac{Gm_im_j}{|\vec{r}_{ij}|},$$

where $\vec{r}_{ij}$ is the vector joining the CM's of the two bodies. We write this as $\vec{r}_{ij} = \vec{d}_{ij} + \vec{x}_i + \vec{x}_j$, where $\vec{d}_{ij}$ is the vector joining their CMs at some initial time and $\vec{x}_{i,j}$ is the displacement of the CM of a given body. We consider the case where s_1 and s_2 are rigid. In applications of interest here, s_1 will be a test mass (e.g. an atom in an interferometer) and s_2 will describe matter gravitationally interacting with s_1 , (e.g. the earth). We will thus assume that a) all constituents of a given body are in a

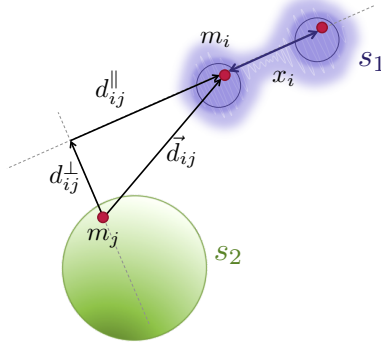


FIG. 2. Test body s_1 in a spatial superposition in the presence of a source mass s_2 . For any pair of elementary masses (m_i, m_j) forming the bodies, the distance $\vec{d}_{ij}$ between them can be decomposed into component $d_{ij}^{\parallel}$ along the direction of the spatial superposition of s_1 and a perpendicular component $d_{ij}^{\perp}$. x_i is marked as a distance between superposed amplitudes of the elementary mass m_i it is, however, the same for all constituents of s_1 .

superposition of equally distant locations (rigidity); b) there is one distinguished direction defined by the superposition of the test mass (while the surrounding matter is well localised), see fig. 2.

It is convenient to write $\vec{d}_{ij} = d_{ij}^{\parallel} \hat{e} + d_{ij}^{\perp} \hat{e}^{\perp}$ where $\vec{x}_i = x_i \hat{e}$ and thus $\hat{e}$ is a unit vector in the direction defined by the superposition and $\hat{e}^{\perp}$ is a unit vector in an orthogonal direction. With the above we can write

$$V_{ij} = \frac{-Gm_i m_j}{\sqrt{(d_{ij}^{\parallel} + x_i + x_j)^2 + (d_{ij}^{\perp})^2}} \approx -Gm_i m_j \times \left(\frac{1}{d_{ij}} - \frac{d_{ij}^{\parallel}}{d_{ij}^3} (x_i + x_j) + \frac{(d_{ij}^{\parallel})^2 - \frac{1}{2}(d_{ij}^{\perp})^2}{d_{ij}^5} (x_i + x_j)^2 \right) \quad (\text{B1})$$

where $d_{ij} = \sqrt{(d_{ij}^{\parallel})^2 + (d_{ij}^{\perp})^2}$.

For any pair (i, j) the “measurement” part of the interaction can thus be taken as $\hat{x}_i \otimes \hat{p}_{m_i} + \hat{x}_j \otimes \hat{p}_{m_j}$ and the “feedback” as $K_{ij} \hat{x}_i \otimes \hat{x}_{m_j} + K_{ij} \hat{x}_j \otimes \hat{x}_{m_i} + \hat{Y}_i \otimes \hat{I}_{m_j} + \hat{Y}_j \otimes \hat{I}_{m_i}$, where $\hat{Y}_{i(j)}$ acts only on mass $m_{i(j)}$. The following master equation for the pair

$$\dot{\rho}_{ij} \approx -\frac{i}{\hbar} [\hat{H}_0 + \hat{Y}_i + \hat{Y}_j + K_{ij} \hat{x}_i \hat{x}_j, \rho_{ij}] - \Gamma_{ij} ([\hat{x}_i, [\hat{x}_i, \rho]] + [\hat{x}_j, [\hat{x}_j, \rho]])$$

is thus obtained, where $\Gamma_{ij} \equiv \frac{1}{4D} + \frac{K_{ij}^2 D}{4\hbar^2}$. Defining $K_{ij} := 2Gm_i m_j \frac{(d_{ij}^{\parallel})^2 - \frac{1}{2}(d_{ij}^{\perp})^2}{d_{ij}^5}$, $Y_i := -Gm_i m_j (\frac{1}{2d_{ij}} - \frac{d_{ij}^{\parallel}}{d_{ij}^3} x_i + \frac{(d_{ij}^{\parallel})^2 - \frac{1}{2}(d_{ij}^{\perp})^2}{d_{ij}^5} x_i^2)$ the master equation describes the induced (approximate) Newtonian interaction between m_i, m_j (since $\hat{Y}_i + \hat{Y}_j + K_{ij} \hat{x}_i \hat{x}_j \approx V_{ij}$, eq. (B1)) with the decoherence term including a gradient of the Newtonian force between the pair, as in eq. (1) but in three dimensions. The dynamics of all $N_1 + N_2$ constituents, described by ρ_{tot} , reads

$$\dot{\rho}_{tot} = -i[\hat{H}_0 + \sum_{i < j}^{N_1 + N_2} V_{ij}, \rho_{tot}] - \sum_{i < j}^{N_1 + N_2} \Gamma_{ij} \left([\hat{x}_i, [\hat{x}_i, \rho_{tot}]] + [\hat{x}_j, [\hat{x}_j, \rho_{tot}]] \right). \quad (\text{B2})$$

Introducing the displacement r_1 (r_2) of the CM of s_1 (s_2), and of the relative degrees of freedom $\hat{x}'_i$ the displacement of any individual constituent can be described by $\hat{x}_i = \hat{r}_1 + \hat{x}'_i$ for $i \leq N_1$ (for constituents of s_1) and $\hat{x}_i = \hat{r}_2 + \hat{x}'_i$ for $i > N_1$ (for constituents of s_2). With the above $[\hat{x}_i, [\hat{x}_i, \rho_{tot}]] = [\hat{r}_1, [\hat{r}_1, \rho_{tot}]] + [\hat{x}'_i, [\hat{x}'_i, \rho_{tot}]] + [\hat{r}_1, [\hat{x}'_i, \rho_{tot}]] + [\hat{x}'_i, [\hat{r}_1, \rho_{tot}]]$, for $i \leq N_1$ and analogously (with $\hat{r}_2$ instead of $\hat{r}_1$) for $i > N_1$. Assuming that the displacements of the relative degrees of freedom are initially uncorrelated with the CM and remain small (both justified by the rigidity of s_1, s_2) and tracing them over results in

following master equation for the CMs of s_1, s_2

$$\begin{aligned} \dot{\rho}_{s_1 s_2} = & -i[\hat{H}_0 + V, \rho_{s_1 s_2}] - 2 \sum_{i < j=1}^{N_1} \Gamma_{ij}[\hat{r}_1, [\hat{r}_1, \rho_{s_1 s_2}]] - 2 \sum_{N_1 < i < j}^{N_1+N_2} \Gamma_{ij}[\hat{r}_2, [\hat{r}_2, \rho_{s_1 s_2}]] \\ & - \sum_{i=1}^{N_1} \sum_{j=N_1+1}^{N_1+N_2} \Gamma_{ij} \left([\hat{r}_1, [\hat{r}_1, \rho_{s_1 s_2}]] + [\hat{r}_2, [\hat{r}_2, \rho_{s_1 s_2}]] \right) \end{aligned} \quad (\text{B3})$$

where $V = \sum_{i < j}^{N_1+N_2} V_{ij} \approx -G \frac{M_1 M_2}{|r_1 - r_2|}$, M_1 and M_2 are correspondingly the total masses of s_1 and s_2 .

Finally, tracing over the degrees of freedom of s_2 one obtains master equation for the CM of s_1 :

$$\dot{\rho}_{s_1} = -i[\hat{H}_0 + V, \rho_{s_1}] - \left(2 \sum_{i < j=1}^{N_1} \Gamma_{ij} + \sum_{i=1}^{N_1} \sum_{j=N_1+1}^{N_1+N_2} \Gamma_{ij} \right) [\hat{r}_1, [\hat{r}_1, \rho_{s_1}]], \quad (\text{B4})$$

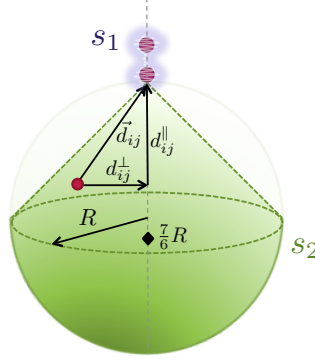


FIG. 3. Estimation of the decoherence effect on a single atom, s_1 , due to Earth: for simplicity we consider only the portion of earth for which the decoherence rate is always greater than the effect stemming from matter concentrated at its centre of mass. This portion of earth's mass is defined as all constituents (atoms) for which $d_{1j}^{\parallel} < d_{1j}^{\perp}$ – inside the cone-and-half-ball, shaded green in the figure. The total mass of the region is 3/4 of the mass of Earth M , and its centre of mass is 7/6 of the Earth's radius R .

We are particularly interested in the case when s_1 is a single atom, $N_1 = 1$, and s_2 is the entire earth. Minimising the decoherence rate for each pair $(1, j)$ gives $\Gamma_{1j}^{\min} = \frac{K_{1j}}{2\hbar}$ and the total decoherence rate is given by $\sum_{j \in \text{earth}} \frac{K_{1j}}{2\hbar}$. Note, that every constituent of the earth acts so as to increase the decoherence rate of s_1 . Here we are looking to relate the resulting decoherence to that given by the earth's CM, as in the original model. Since K_{1j} as a function of $(d_{1j}^{\parallel}, d_{1j}^{\perp})$ is convex only for $|d_{1j}^{\parallel}| < |d_{1j}^{\perp}|/\sqrt{2}$ we consider only a portion $\mathcal{C}$ of the earth's mass, which lies within the volume where K_{1j} is convex, see fig. 3.

The overall decoherence rate is greater than that stemming from particles in $\mathcal{C}$, which itself is greater than decoherence coming from the centre of mass of $\mathcal{C}$. Since $|d_{1j}^{\parallel}| < |d_{1j}^{\perp}|/\sqrt{2} < |d_{1j}^{\perp}|$, the region $\mathcal{C}$ can be taken as a cone of height R and support of area πR^2 together with a half ball of radius R . Assuming a constant density for body M_2 , the mass of $\mathcal{C}$ is $\frac{3}{4}M_2$ and its CM is at a distance $\frac{7}{6}R$ from the top surface, as depicted in fig. 3. The quantity

$$\Gamma_{M_2;R}^{\min} = \frac{3}{4} \left(\frac{6}{7} \right)^3 \Gamma_{KTM}(M_1, M_2, R) = 0.47 \Gamma_{KTM}(M_1, M_2, R)$$

where $\Gamma_{KTM}(M_1, M_2, R) = \frac{GM_1 M_2}{R^3}$ is the lower bound on the decoherence rate of mass M_1 due to the homogeneous ball of mass M_2 and radius R . Note that in the above the $\Gamma_{KTM}(M_1, M_2, R)$ is the decoherence rate as per the original KTM model for masses M_1, M_2 at a distance R .

Appendix C: Torsion Balance

Here we consider an application of the KTM model to torsion balance experiments. These experiments measure the gravitational constant G by balancing the torque produced by the gravitational attraction between massive objects situated on the balance. We show here how the measurement/feedback model can be used to simulate a classical potential in this context.

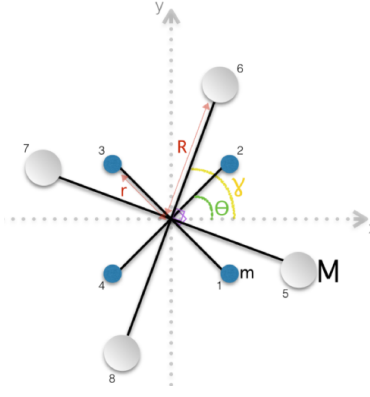


FIG. 4. Set-up of the torsion balance experiment.

The Hamiltonian of the system is

$$H = \sum_i \frac{1}{2} m_i (\dot{x}_i^2 + \dot{y}_i^2) + \sum_{i,j} V_{ij}, \quad (C1)$$

where i runs over all (spherical) bodies in the experiment and V_{ij} denotes the (Newtonian) gravitational potential between pairs of bodies, with each body regarded as a point-like object located at its centre-of-mass.

The actual experiment consists of 8 bodies, 4 identical small bodies each of mass m , and 4 other identical large bodies each of mass M , as illustrated in figure 4. Since all bodies in the experiment are in the same plane, we can write the above hamiltonian as

$$\begin{aligned} H &= 2m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right) + 2M \left(\dot{R}^2 + R^2 \dot{\gamma}^2 \right) + \sum_{i,j} V_{ij} \\ &= 2mr^2 \dot{\theta}^2 + 2MR^2 \dot{\gamma}^2 + \sum_{i < j} V_{ij} \end{aligned} \quad (C2)$$

in polar coordinates (r, θ) and (R, γ) for the small and large bodies respectively, where the latter relation follows from the rigidity of the balance arms. The setup of the experiment ensures that V_{ij} depends only on the variable $\alpha = \gamma - \theta$, and so

$$\begin{aligned} H &= \frac{mr^2 + MR^2}{4mr^2 MR^2} p_\alpha^2 + \frac{p_\xi^2}{4(mr^2 + MR^2)} + \frac{4GmM}{\sqrt{r^2 + R^2 + 2rR \cos \alpha}} + \frac{4GmM}{\sqrt{r^2 + R^2 + 2rR \sin \alpha}} \\ &\quad + \frac{4GmM}{\sqrt{r^2 + R^2 - 2rR \cos \alpha}} + \frac{4GmM}{\sqrt{r^2 + R^2 - 2rR \sin \alpha}} + \frac{4Gm^2}{\sqrt{2}r} + \frac{4GM^2}{\sqrt{2}R} + \frac{Gm^2}{r} + \frac{GM^2}{R} \end{aligned} \quad (C3)$$

where p_α is the conjugate momentum to α and p_ξ is the conjugate momentum to the variable $\xi \equiv \frac{mr^2 \theta + MR^2 \gamma}{mr^2 + MR^2}$.

The relevant variable is small deviations of the angle α away from its equilibrium value α_0 . Writing $\alpha = \alpha_0 + \delta\alpha$, where $\delta\alpha \ll \alpha_0$, the Hamiltonian can be approximated as

$$H = \frac{p_\alpha^2}{2I_{eff}} + \frac{p_\xi^2}{4(mr^2 + MR^2)} + B\delta\alpha + C(\delta\alpha)^2 + \dots \quad (C4)$$

where $I_{eff} = \frac{2mr^2 MR^2}{mr^2 + MR^2}$ is the reduced moment of inertia and

$$\begin{aligned} B &= \sum_{n=1}^3 \frac{4GmMrR \sin(\alpha_0 + (n+1)\pi/2)}{(r^2 + R^2 - 2rR \sin(\alpha_0 + n\pi/2))^{\frac{3}{2}}} \\ C &= - \sum_{n=1}^3 \frac{2GmMrR \sin(\alpha_0 + n\pi/2)}{(r^2 + R^2 - 2rR \sin(\alpha_0 + n\pi/2))^{\frac{3}{2}}} + \sum_{n=1}^3 \frac{6GmMr^2 R^2 \sin^2(\alpha_0 + (n+1)\pi/2)}{(r^2 + R^2 - 2rR \sin(\alpha_0 + n\pi/2))^{\frac{5}{2}}} \end{aligned} \quad (C5)$$

and we have dropped the irrelevant constant terms from (C3).

In the context of the general master equation (1), we have $\hat{x} \rightarrow \delta\hat{\alpha} \equiv \Delta\alpha$, and $K = 2C$. Writing $D = \frac{\hbar}{2C\epsilon}$, we obtain

$$\dot{\rho} = -\frac{i}{\hbar} \left[\frac{\hat{p}_\alpha^2}{2I_{eff}} + \frac{\hat{p}_\xi^2}{4(mr^2 + MR^2)} + B\delta\hat{\alpha} + C\delta\hat{\alpha}^2, \rho \right] - \frac{C}{2\hbar} \left(\epsilon + \frac{1}{\epsilon} \right) [\delta\hat{\alpha}, [\delta\hat{\alpha}, \rho]] \quad (C6)$$

To empirically constrain the KTM model using this class of experiments, we adopt the perspective that world measurements on Newton's constant G [45] have a certain degree of scatter, and that this scatter can provide a bound on how large the gravitational repeated-measurement-feedback effect from (C6) is. We have assumed that the bodies are pointlike, but as we have seen from earlier sections taking into account the finite size of the bodies will lead to corrections in D (or Γ) that are of order unity.

For the Cavendish experiment the total error in G is

$$\frac{\Delta G}{G} = \frac{\delta(\Delta\alpha)}{\Delta\alpha} + \frac{\delta k}{k} - \left(\frac{\delta M}{M} + \frac{\delta m}{m} + 4\frac{\delta r_a}{r_a} + 4\frac{\delta r_b}{r_b} - 5\frac{\delta R_{ac}}{R_{ac}} - 5\frac{\delta R_b}{R_b} + \alpha_{CT} \right) + \frac{\delta\tau_c}{\tau_c} \quad (C7)$$

where the meaning of these various quantities is explained in ref. [44] (see equation (11.10) of [44]). The minimal constraint on $\Delta\alpha$ is therefore

$$\left| \frac{\delta(\Delta\alpha)}{\Delta\alpha} \right| \leq \left| \frac{\Delta G}{G} \right|$$

where $\Delta\alpha = |\langle(\delta\hat{\alpha})^2\rangle|$ and $\delta(\Delta\alpha) = \sqrt{\langle(\delta\hat{\alpha})^2\rangle}$, the latter quantity being the time-averaged variance computed from the repeated measurement process. This quantity is

$$\langle(\delta\hat{\alpha})^2\rangle = \frac{\hbar}{8I_{eff}} T \left(\epsilon + \frac{1}{\epsilon} \right)$$

where T is the timescale over which the experiment is performed.

To estimate the size of $\delta(\Delta\alpha)$, we have $m = 1.2$ kg, $M = 11$ kg, $r = 120$ mm, $R = 214$ mm, and $\alpha_0 = 18.9^\circ$, yielding $I_{eff} = 8.35 \times 10^{-3}$ kg m². Hence

$$\langle(\delta\hat{\alpha})^2\rangle = 1.58 \times 10^{-33} T \left(\epsilon + \frac{1}{\epsilon} \right) \text{ s}^{-1}$$

Consequently

$$\begin{aligned} \left| \frac{\delta(\Delta\alpha)}{\Delta\alpha} \right| &= \sqrt{1.58 \times 10^{-33} T \left(\epsilon + \frac{1}{\epsilon} \right)} \leq \left| \frac{\Delta G}{G} \right| \sim 10^{-6} \\ \Rightarrow T \left(\epsilon + \frac{1}{\epsilon} \right) &\leq \sim 6 \times 10^{20} \end{aligned} \quad (C8)$$

for $\Delta\alpha \sim 1$ radian. For an experiment on the order of 1 day = $3600 \times 24 = 86400$ seconds, then

$$\left(\epsilon + \frac{1}{\epsilon} \right) \leq \sim 7 \times 10^{15}$$

and the decoherence rate satisfies

$$\frac{C}{2\hbar} \left(\epsilon + \frac{1}{\epsilon} \right) \leq \sim 3.7 \times 10^{40}.$$

A more sophisticated treatment involving the compositeness of the bodies in the torsion balance will provide corrections of order unity to C . Since the above constraint is much weaker than that provided by the fountain experiments we shall not pursue this any further.

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